Joint Model Importance via the Rashomon Set

Behavioural Comparison of Near-Equally-Good Models

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Beyond the single best model

- In tabular ML, many models reach indistinguishable predictive performance.
- For a tolerance $\varepsilon > 0$, the *Rashomon set*

$$\mathrm{RS}(\varepsilon) \;=\; \left\{\, \hat{f}_{\mathcal{D},\boldsymbol{\lambda}} \;:\; \mathcal{R}(\hat{f}_{\mathcal{D},\boldsymbol{\lambda}}) \le \mathcal{R}(\hat{f}_{\mathcal{D},\boldsymbol{\lambda}^*}) \cdot (1+\varepsilon),\;\; \boldsymbol{\lambda} \in \Lambda \,\right\}$$

  collects every model within ε of the best.

  Notation: $\hat{f}_{\mathcal{D},\boldsymbol{\lambda}}$ = model trained on data $\mathcal{D}$ with hyperparameter configuration $\boldsymbol{\lambda} \in \Lambda$; $\boldsymbol{\lambda}^*$ = risk-minimizing configuration; $\mathcal{R}(\cdot)$ = risk.
- Members of $\mathrm{RS}(\varepsilon)$ can disagree substantially on individual predictions and on feature importance.
- **Question.** How are these equally good models related to one another, and what does the collective tell us beyond the single best?

Pipeline

1. For every $\hat{f}_{\mathcal{D},\boldsymbol{\lambda}} \in \mathrm{RS}(\varepsilon)$, compute the prediction vector $\hat{\mathrm{y}}$ on a held-out 2/3 validation split.
2. Build the pairwise *behavioural distance* matrix D with entries

$$d_{ij} \;=\; \|\hat{\mathrm{y}}_i - \hat{\mathrm{y}}_j\|_p \;=\; \Bigl(\sum_{o=1}^{n_{\mathrm{val}}} \bigl|\hat{y}_{i,o} - \hat{y}_{j,o}\bigr|^p\Bigr)^{1/p},$$

   where i, j index models in $\mathrm{RS}(\varepsilon)$ and o runs over the n_{val} validation observations.
3. View D three ways:
   - hierarchical clustering (`hclust`)
   - classical MDS (`cmdscale`, embedding dimension $k = 2$)
   - partitioning around medoids (`cluster::pam`)

Distances computed for both Euclidean ($p = 2$) and Manhattan ($p = 1$); results shown are Euclidean (Manhattan tells the same story).

Setup

Rashomon tolerance $\varepsilon = 1\%$ of the minimum risk.
Five learner families considered for every task: `tree`, `glmnet`, `xgb`, `nnet`, `svm`.

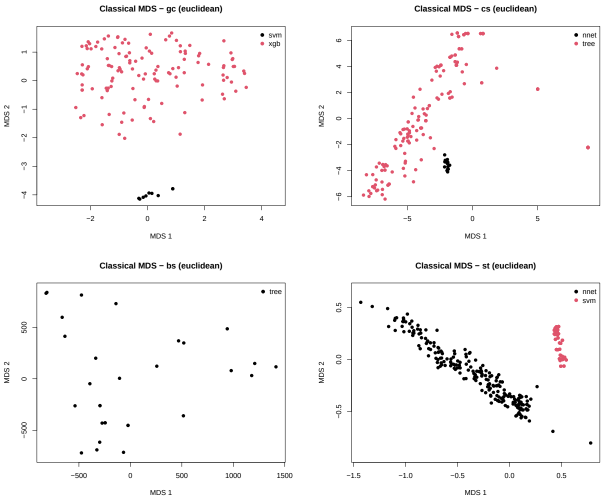
Key	Task	Type
`gc`	German Credit	classification
`cs`	COMPAS	classification
`bs`	Bike Sharing	regression
`st`	Synthetic	regression

Who survives the Rashomon cut?

Task	Surviving families (n models)
`gc`	`svm` (9), `xgb` (125)
`cs`	`nnet` (15), `tree` (261)
`bs`	`tree` (30)
`st`	`nnet` (216), `svm` (160)

First finding. At $\varepsilon = 1\%$ only one or two learner families survive per task — *learner choice* is the first dimension of disagreement inside $\mathrm{RS}(\varepsilon)$.

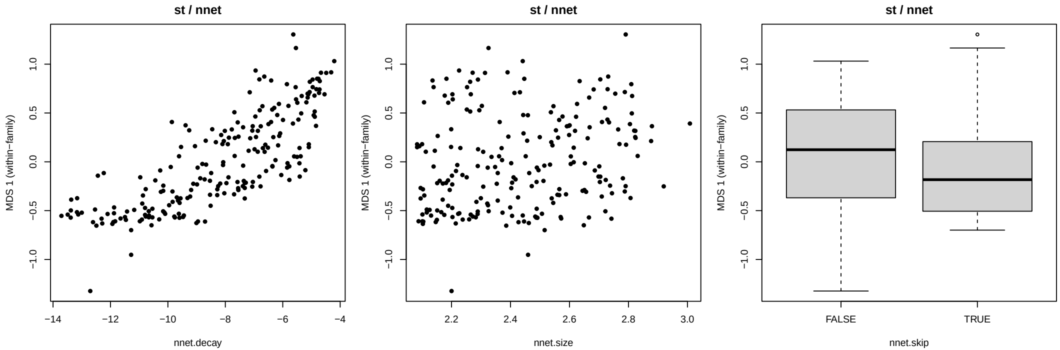
MDS view — all four tasks



Reading the MDS plots

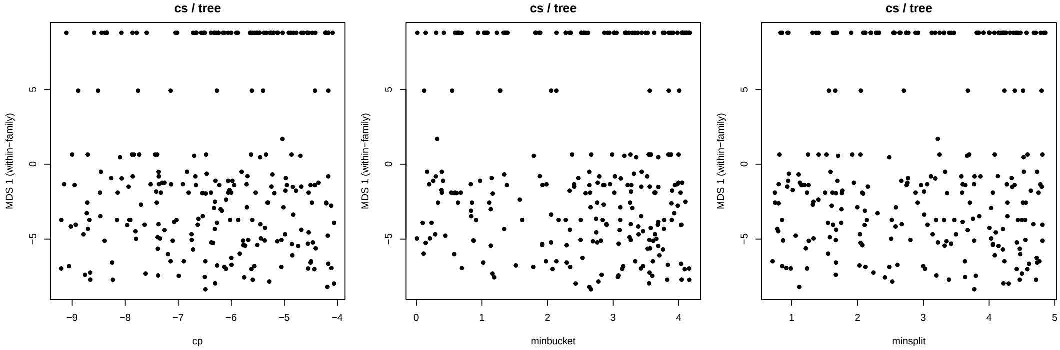
- Wherever two families coexist, behavioural distance **separates them cleanly** — the dominant axis is family identity, not within-family variation.
- Within-family variance is *asymmetric*: `svm` and `nnet` form tight clusters; `tree` and `xgb` scatter widely despite fewer tunable hyperparameters.
- Several families trace out a **1D-manifold** in the MDS plane (e.g. `st/nnet`, `cs/tree`) — next two slides.

Within-family structure I — st / nnet



- Each panel: 1D within-family MDS coordinate vs. one hyperparameter.
- The within-family manifold **is the `decay` axis** — near-monotone mapping.
- `size` and `skip` have no effect on behavioural position.

Within-family structure II — cs / tree

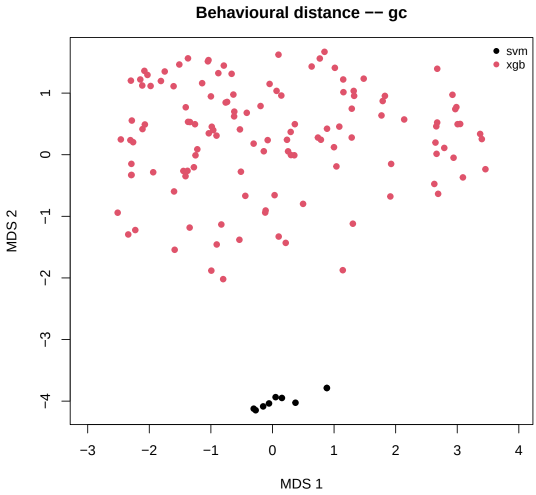


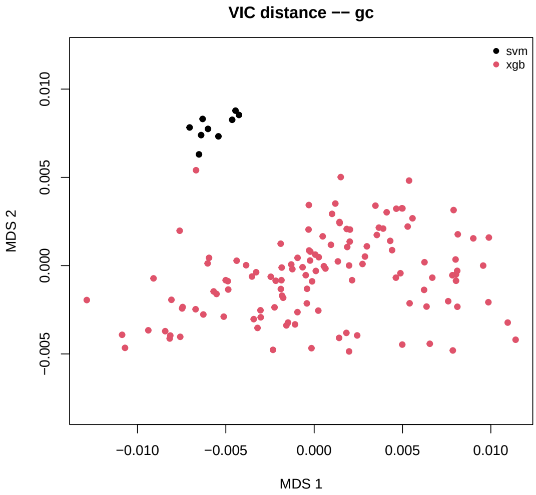
- Within-family MDS coordinate vs. `cp`, `minbucket`, `minsplit`.
- Trees on COMPAS are **bimodal**: a tight cluster at MDS 1 ≈ 8 and a diffuse population below zero.
- *No single tuned hyperparameter* separates the two modes. **Speculation:** a discrete behavioural split driven by structure (e.g. root split, depth) that hyperparameter tuning does not reach — to be confirmed.

Scope — behaviour is not importance

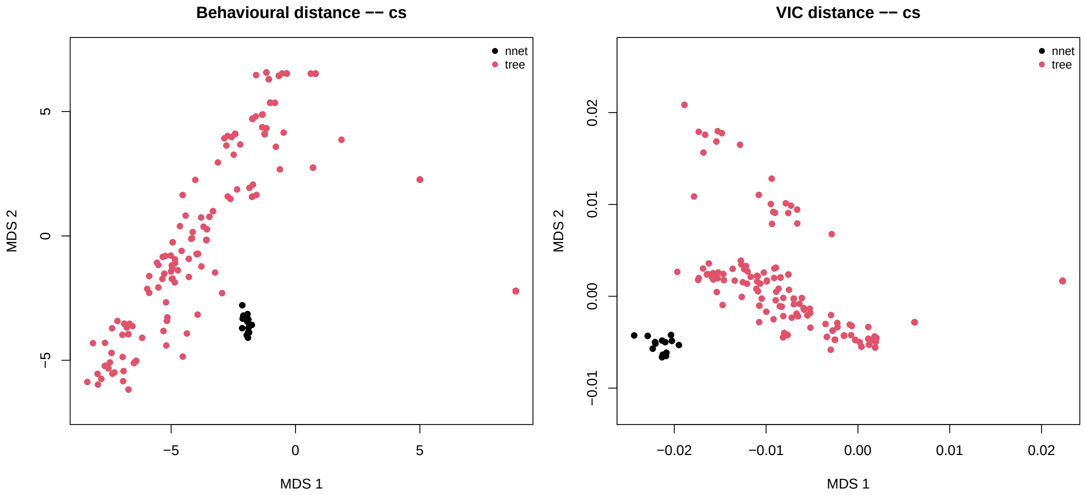
- **What we measure:** behavioural distance $d_{ij}^{\text{beh}} = \|\hat{y}_i - \hat{y}_j\|_p$ from validation predictions.
- **What "joint model importance" asks:** how do per-model importance vectors $\phi^{(i)}$ vary across $i \in \mathrm{RS}(\varepsilon)$? (Variable-Importance Cloud, VIC.)
- **Independent in principle:** identical predictions need not imply identical importance (and vice versa).
- **Compared empirically next slide:** same models, same colouring, behavioural distance left, VIC distance right.

Behavioural vs VIC — gc

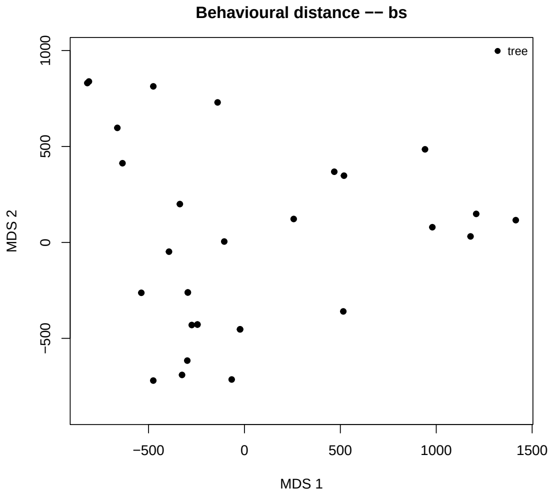


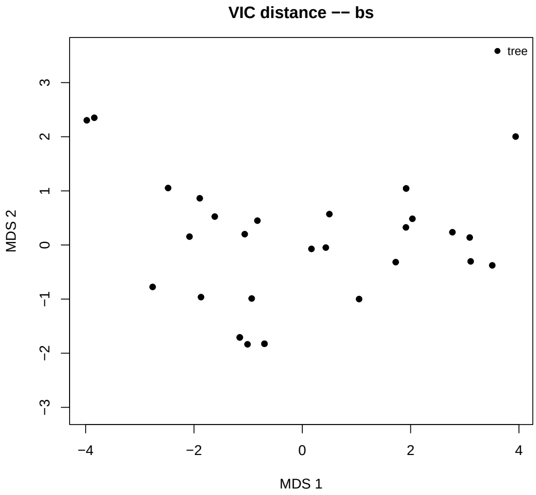


Behavioural vs VIC — cs

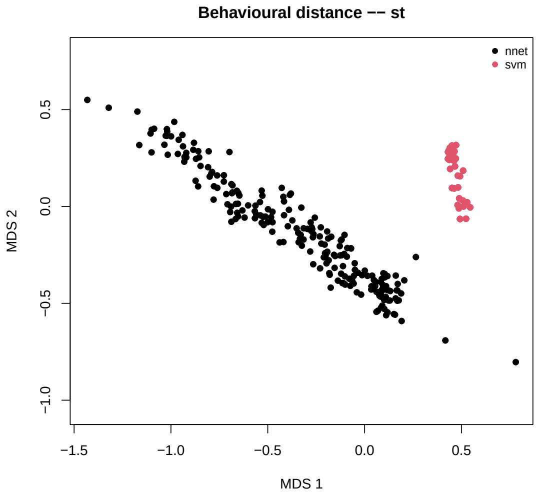


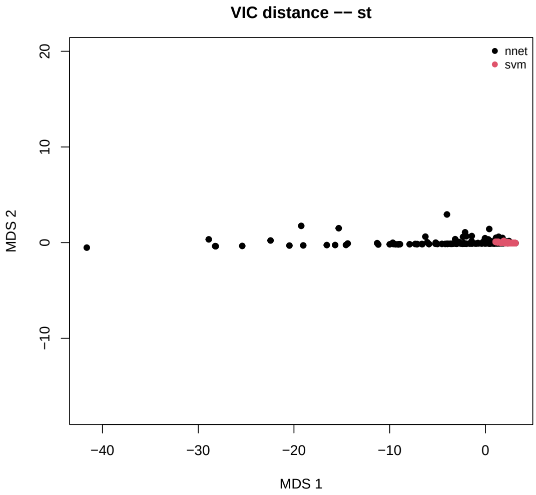
Behavioural vs VIC — bs





Behavioural vs VIC — st





Behavioural vs VIC — what we see

- `gc`: family separation largely preserved in both views, though a handful of `xgb` points overlap the `svm` cluster in VIC.
- `cs`: family separation preserved, but the within-family behavioural bimodality of trees (Slide 9) **vanishes** on the importance side.
- `bs`: single family in $\mathrm{RS}(\varepsilon)$ — uninformative.
- `st`: behavioural family separation **collapses** in VIC — `nnet` and `svm` share the same importance profile despite predicting very differently.
- **Take-away:** behavioural clustering is *not* a reliable proxy for importance clustering. The disagreement is empirical and *task-dependent*.

First interpretation

- Within $\mathrm{RS}(\varepsilon)$, **behaviour has two-level structure:** learner family + position on the family's behavioural manifold.
- Across every task observed, family choice dominates behavioural variance — not hyperparameter setting.
- The within-family manifold can be smooth and HP-aligned (`st`/`nnet` $\sim$ `decay`) or discrete and HP-orthogonal (`cs`/`tree`; *causal mechanism still speculative*).
- **But behaviour is not importance:** family separation only partly survives in VIC (`gc`, `cs`), and the within-family discreteness of `cs`/`tree` vanishes entirely. For `st`, the families themselves collapse. *Joint model importance therefore requires both views.*

Next steps

- ε **sensitivity.** How do behavioural and VIC clusters shift as the Rashomon tolerance opens?
- **Per-feature analysis.** Which features drive the behaviour-vs-VIC disagreement? Especially for the `st` collapse.
- **Choose the number of clusters** k **for PAM via silhouette**, then read out the medoids as "representative" members of $\mathrm{RS}(\varepsilon)$.
- **Disagreement map.** Where in feature space do Rashomon-set models disagree the most?
- **Discrete tree mode on** `cs`**.** Is the behavioural bimodality driven by structural differences (root split, depth) that hyperparameters do not capture?

Thank you.

Questions?

Backup — the behavioural distance matrix in detail

Stack the Rashomon-set prediction vectors row-wise into a prediction matrix:

$$\mathrm{P} = \begin{pmatrix} \hat{y}_{1,1} & \hat{y}_{1,2} & \cdots & \hat{y}_{1,n_{\text{val}}} \\ \hat{y}_{2,1} & \hat{y}_{2,2} & \cdots & \hat{y}_{2,n_{\text{val}}} \\ \vdots & \vdots & & \vdots \\ \hat{y}_{n,1} & \hat{y}_{n,2} & \cdots & \hat{y}_{n,n_{\text{val}}} \end{pmatrix} \in \mathbb{R}^{n \times n_{\text{val}}}$$

with one row per model in $\mathrm{RS}(\varepsilon)$ (here $n = |\mathrm{RS}(\varepsilon)|$) and one column per held-out validation observation (here n_{val}).

Row-wise L_p distances collapse the observation axis and yield the symmetric matrix

$$\mathrm{D} \in \mathbb{R}^{n \times n}, \qquad d_{ij} = \|\hat{\mathrm{y}}_i - \hat{\mathrm{y}}_j\|_p.$$

Rows / columns of D now index *models*, not observations. Hence **one MDS point = one model**; the cloud sits in $\mathbb{R}^n$ and is visualised in $\mathbb{R}^2$ via `cmdscale`.

Backup — the three views of D

- **Hierarchical clustering** (`hclust`): bottom-up greedy merging of the two closest clusters; output is a dendrogram. Cluster count is fixed by where one cuts.
- **Classical MDS** (`cmdscale`, $k = 2$): eigendecomposition of the double-centred squared distance matrix, top-k eigenvectors as coordinates. Each model becomes a point in $\mathbb{R}^k$ whose pairwise Euclidean distances approximate D. For Euclidean D **equivalent to PCA** up to sign and rotation.
- **Partitioning around medoids** (`cluster::pam`): pick k medoids (representative models) minimising the total distance from each model to its nearest medoid. Robust analogue of k-means for arbitrary distance matrices; best k via average silhouette width.